Akoizo Original Practice - 2027 Rotation Form 4

Division C | Circuit Lab | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) XOR(1,1) equals…

Unit: Digital logic and combinational circuits.
Worked-case data: For A=1, B=1, find half-adder sum and carry.

A. 0

B. 1

C. 2

D. Undefined

Answer: _____

2. (2 points) Two 100 Ω resistors in series give…

Unit: Diagrams, nodes, series, parallel, and Ohm's law.
Worked-case data: Find Req for 100 Ω and 300 Ω in parallel on 6 V.

A. 200 Ω

B. 50 Ω

C. 100 Ω

D. 10000 Ω

Answer: _____

3. (2 points) An ohmmeter generally requires…

Unit: Breadboards, multimeters, and a first virtual build.
Worked-case data: A known 100 Ω resistor has 2 V across it; an unknown in series has 4 V. Find the unknown.

A. A live mains circuit

B. Maximum possible voltage

C. No instrument instructions

D. A de-energized suitable circuit

Answer: _____

4. (2 points) A multiplexer…

Unit: Digital logic and combinational circuits.
Worked-case data: For A=1, B=1, find half-adder sum and carry.

A. Always adds inputs

B. Stores a clocked bit by definition

C. Measures resistance

D. Selects an input

Answer: _____

5. (4 points) Why trace nodes first?

Unit: Diagrams, nodes, series, parallel, and Ohm's law.
Worked-case data: Find Req for 100 Ω and 300 Ω in parallel on 6 V.

Answer: _____
_____
_____

6. (4 points) Why check rail continuity?

Unit: Breadboards, multimeters, and a first virtual build.
Worked-case data: A known 100 Ω resistor has 2 V across it; an unknown in series has 4 V. Find the unknown.

Answer: _____
_____
_____

7. (4 points) Why test all input combinations?

Unit: Digital logic and combinational circuits.
Worked-case data: For A=1, B=1, find half-adder sum and carry.

Answer: _____
_____
_____

8. (4 points) Why does divider loading matter?

Unit: Diagrams, nodes, series, parallel, and Ohm's law.
Worked-case data: Find Req for 100 Ω and 300 Ω in parallel on 6 V.

Answer: _____
_____
_____

9. (4 points) Why isolate an unknown for resistance measurement?

Unit: Breadboards, multimeters, and a first virtual build.
Worked-case data: A known 100 Ω resistor has 2 V across it; an unknown in series has 4 V. Find the unknown.

Answer: _____
_____
_____

10. (4 points) What does a Boolean model omit?

Unit: Digital logic and combinational circuits.
Worked-case data: For A=1, B=1, find half-adder sum and carry.

Answer: _____
_____
_____

11. (2 points) A 10 V ideal battery drives a 5 ohm resistor. Find current.

Answer: _____
_____
_____

12. (2 points) A series pair has resistances 6 and 8 ohms. Find equivalent resistance.

Answer: _____
_____
_____

13. (2 points) Two identical 12 ohm resistors are placed in parallel. Find equivalent resistance.

Answer: _____
_____
_____

14. (2 points) An ideal resistor carries 2 A at 10 V. Find power.

Answer: _____
_____
_____

15. (2 points) A node has incoming currents 7 A and 2 A and one outgoing current. Find the outgoing current.

Answer: _____
_____
_____

16. (2 points) Explain why an ideal voltmeter is connected in parallel rather than used as a series current path.

Answer: _____
_____
_____

17. (2 points) Give the output of an AND gate for inputs 1 and 0, and explain.

Answer: _____
_____
_____

18. (2 points) A silicon diode is reverse biased below breakdown. Contrast its ideal current with a forward-biased conducting diode.

Answer: _____
_____
_____

19. (2 points) A closed DC loop has a 9 V source and a 4 V resistor drop. Find the remaining series resistor drop and state the governing law.

Answer: _____
_____
_____

20. (2 points) Two identical LEDs must be equally bright in a low-voltage teaching circuit. Explain why their branch currents and current-limiting arrangements matter.

Answer: _____
_____
_____